



ZHIMAI JING-ZHANG

AI-Native Innovation Belt - Concept Proposal

Zhimai Jing-Zhang - An AI-Native Innovation Belt

Century-old Jing-Zhang Railway Heritage Park x AI-Native Innovation x Urban Design

Hermes Agent (FTARCH) - Wengyongsheng - 2026

This proposal is a concept/reference plan for professional team further study, not a substitute for formal planning

1. Design Basis and Core Concept

This proposal is based on the open-city.ai public task package, the Beijing Municipal Planning and Natural Resources Commission Haidian Branch pre-qualification announcement (2026-05-09), and the provisional boundary provided by the repository maintainer. Core concept **'Zhimai Jing-Zhang'**: Reading the Jing-Zhang Railway Heritage Park as the main artery of AI-native innovation - the linear logic of the century-old railway (connection, transmission, nodes, iteration) is translated into the flowing skeleton of data, computing power, talent and scenarios in the AI era. The three cores (Zhongzhi Park, AI Origin, Dazhongsi) are synaptic nodes on the pulse, and the two wings (Science & Technology Service Wing, Scenario Empowerment Wing) provide element support and scenario validation.

Spatial Framework: Three Cores, One Axis, Two Wings. Three cores: Zhongzhi Park AI Independent Innovation Acceleration Zone (191.9ha), Beijing AI Origin Community (104.3ha), Dazhongsi AI Industry Agglomeration Zone (72.4ha); One axis: Jing-Zhang Green Corridor public main artery; Two wings: Zhongguancun Science & Technology Service Wing (West) and Xiaoyue River Scenario Empowerment Wing (East). Overall research scope 43.6km², concept design scope 11.4km².

1909 Protocol Railway signal block system -> spatial interaction protocol: Block section (sandbox) - Signal (state visible) - Interlocking (safety linkage) - Token (encrypted credential) - Dispatch (human-in-the-loop) - Switchback (routing)	AI-Native Spatial Typology Computing Water Tower (edge computing landmark) Prompt Plaza (human-AI co-creation) Model Engine Room (visible training) Switch Plaza (democratic decision-making)	Spatial Version Control City OS supports AI configuration branch-test-merge-rollback (like git), community can trial-and-error, rollback, fork different AI configurations
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11.4 km² Design Scope	368.6 ha Three Core Areas	31.0% Green Space Ratio	441 bldgs Concept Buildings	15 sites AI Scenario Nodes	4 sites Pilgrimage Landmarks
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Spatial Structure: One Axis, Three Cores, Two Wings. **One Axis:** Jing-Zhang Green Corridor main artery ~9km, from North 5th Ring to Beijing North Station, AI-native innovation main artery. **Three Cores:** Zhongzhi Park (North Core 191.9ha, AI full-stack independent innovation acceleration zone), AI Origin Community (Central Core 104.3ha, 24h mixed innovation community), Dazhongsi (South Core 72.4ha, AI industry agglomeration zone). **Two Wings:** West Wing Science & Technology Service (298.7ha), East Wing Scenario Empowerment (338.7ha). Overall research scope 43.6km², design scope 11.4km².

Six Agent Task Coverage: 1) Overall concept and function coordination 2) AI full-stack independent innovation system 3) AI+ scenario empowerment (15 scenario cards/4 test verification/5 user personas) 4) AI public space and pilgrimage landmarks (4 sites) 5) Cultural narrative (from herringbone to network) 6) Global AI innovation events and long-term operations.

2. Spatial Overview

Fig.1 Overview - Three Cores + One Axis + Two Wings spatial structure (provisional boundary, concept proposal)

3. Current Baseline and Real Data Anchors

Proposal Positioning: Overlay AI-native layer on built urban base, not greenfield development. Jing-Zhang Railway Heritage Park Phase II opened on August 6, 2026, forming a 9km, 53ha continuous linear park, directly serving 70 communities with ~450,000 residents.

Dimension	Key Data	Source
Jing-Zhang Green Belt	9km/53ha/70 communities/450k people (Phase II opened Aug 2026)	Beijing Daily
AI Enterprises	1900+, 51 unicorns (40%+ of city), full industry chain	Haidian 2024 Stats
AI Talent	12.3k scholars (80%+ of city), 646 academicians (35.8% national), 2M talent	Haidian 2024 Stats
Large Models	104 registered (70% of city), 4 of 'Six Tigers' within 2km	2025 Press Conf
Industry Scale	AI core industry 282.2B yuan (2024, 80% city, +30%), 10kP computing	2025 Press Conf
Economy R&D	GDP 1.29T (1/4 city), R&D intensity 12%+, per capita disposable 106k	Haidian 2024 Stats
Demographics	Permanent pop 3.12M, 60+ 23%, 0-14 11.8%, migrant 1.04M	Haidian 2024 Stats
Universities	92 national key labs (63% city/18% national), 37 universities	Haidian 2024 Stats
Transit	Line 13 along corridor, Zhichun Rd 310k/day, Line 18 capacity release	Beijing Transit
Official Plan	Two zones-one belt-multi points: Wudaokou+Dazhongsi pilot+Jing-Zhang belt (2024)	Haidian Gov

Core Area	Real Anchors (public data)	Proposal Response
Zhongzhi Park (N)	BUAA AI Institute/Embodied AI Robot Institute, CAICT, Railway Academy, 8-university consortium	AI full-stack innovation acceleration: training pilot + red team test + computing sandbox
AI Origin (C)	Tsinghua/PKU, CAS Institutes of Automation/Software/Physics, 200+ AI enterprises, 92 new registrations	24h mixed innovation community: researcher-entrepreneur-resident symbiosis + Origin Monument
Dazhongsi (S)	BUPT/BNU/BJTU, official AI district pilot, Line 13/12 transfer	AI+ consumption/legal/finance scenarios + Global AI Milestone Pavilion

4. Land Use Structure: Three Cores, One Axis, Two Wings

Land use zones cover the overall design scope (11.4km²) without overlap: three cores as functional (368.6ha), Jing-Zhang Green Corridor as north-south public main artery, two wings as urban innovation hinterland. All land use is concept proposal, not statutory.

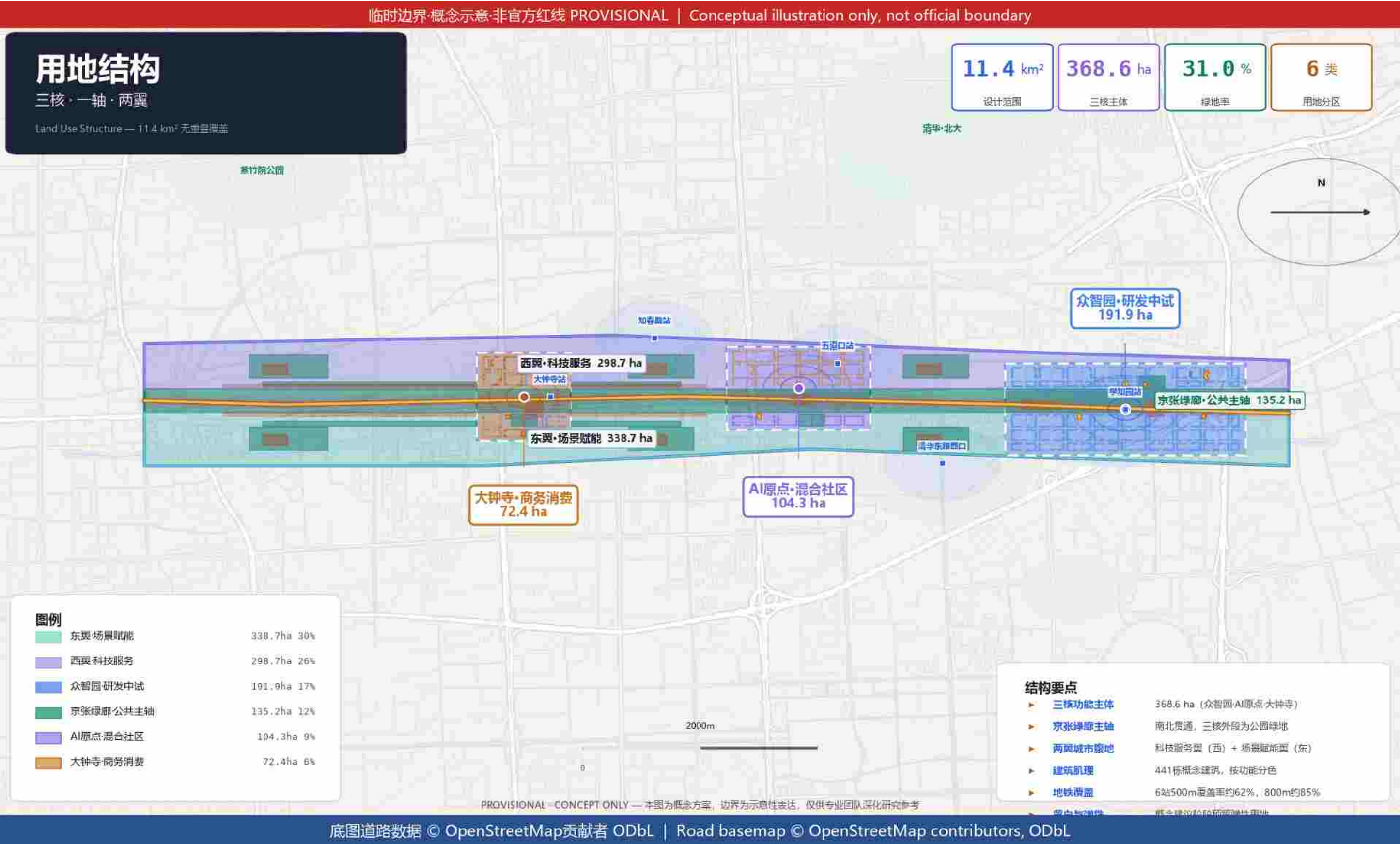


Fig.2 Land Use - Three Cores, One Axis, Two Wings, covering 11.4km² without overlap

5. Three Cores Detailed Design

Fig.3 Three Core Areas - Zhongzhi Park (N) / AI Origin Community (C) / Dazhongsi (S)

ID	Name	Location	Concept
LM-01	Zhimai Origin Monument	AI Origin Community Central Plaza	Marks AI-native innovation starting point
LM-02	Open Source Achievement Gallery	Zhongzhi Park entrance old factory	China open source AI achievement hall of honor
LM-03	Developer Promenade	Middle section of green corridor	Outstanding developer code starlight
LM-04	Global AI Milestone Pavilion	Dazhongsi gateway plaza	Global AI development node display

Concept Section and Skyline Intention

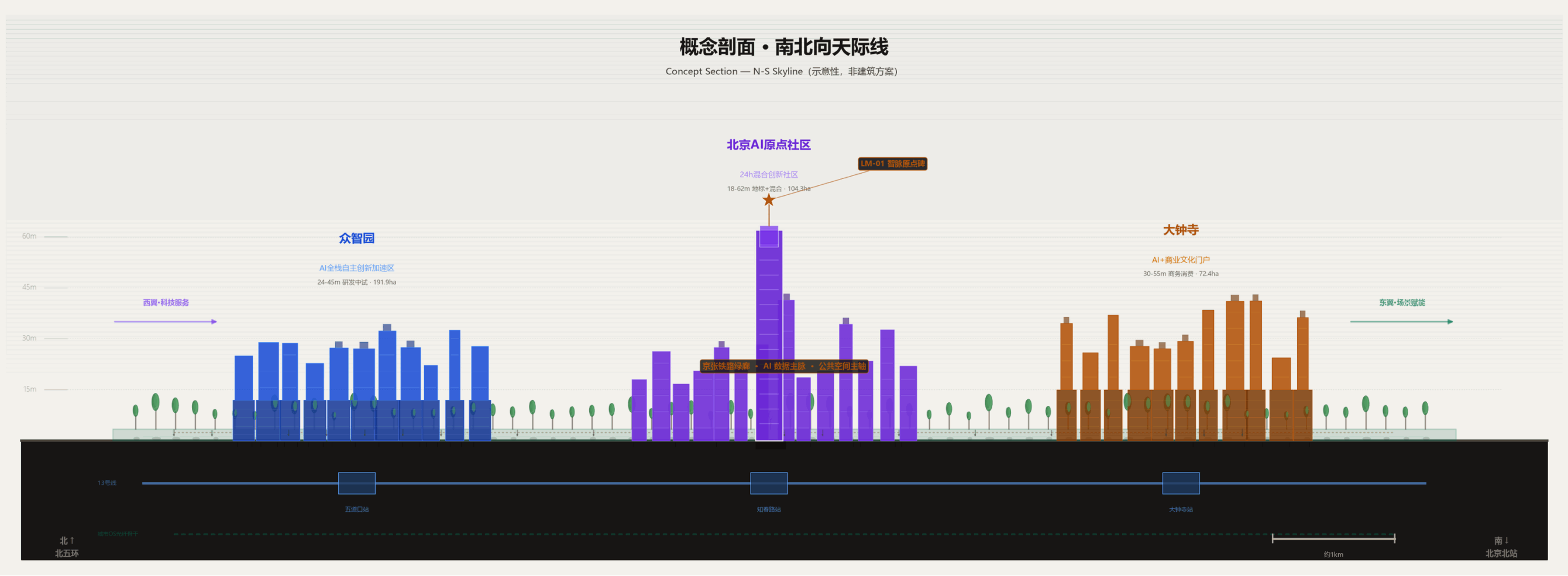


Fig.4 Concept Section - North-south skyline and three core height intention (not architectural scheme, concept proposal)

Spatial Height Intention: Zhongzhi Park focuses on R&D; pilot, concept height 24-45m, preserving industrial heritage texture; AI Origin Community is mixed function, concept height 18-60m, with landmark tower and LM-01 Zhimai Origin Monument at center; Dazhongsi focuses on business consumption, concept height 30-60m. Jing-Zhang Green Corridor is continuous ground park layer, golden dashed line marks century-old railway heritage alignment. This section is spatial intention expression, **not architectural scheme**; building height, volume and layout are subject to professional team further study and regulatory approval.

Urban Design Guidelines

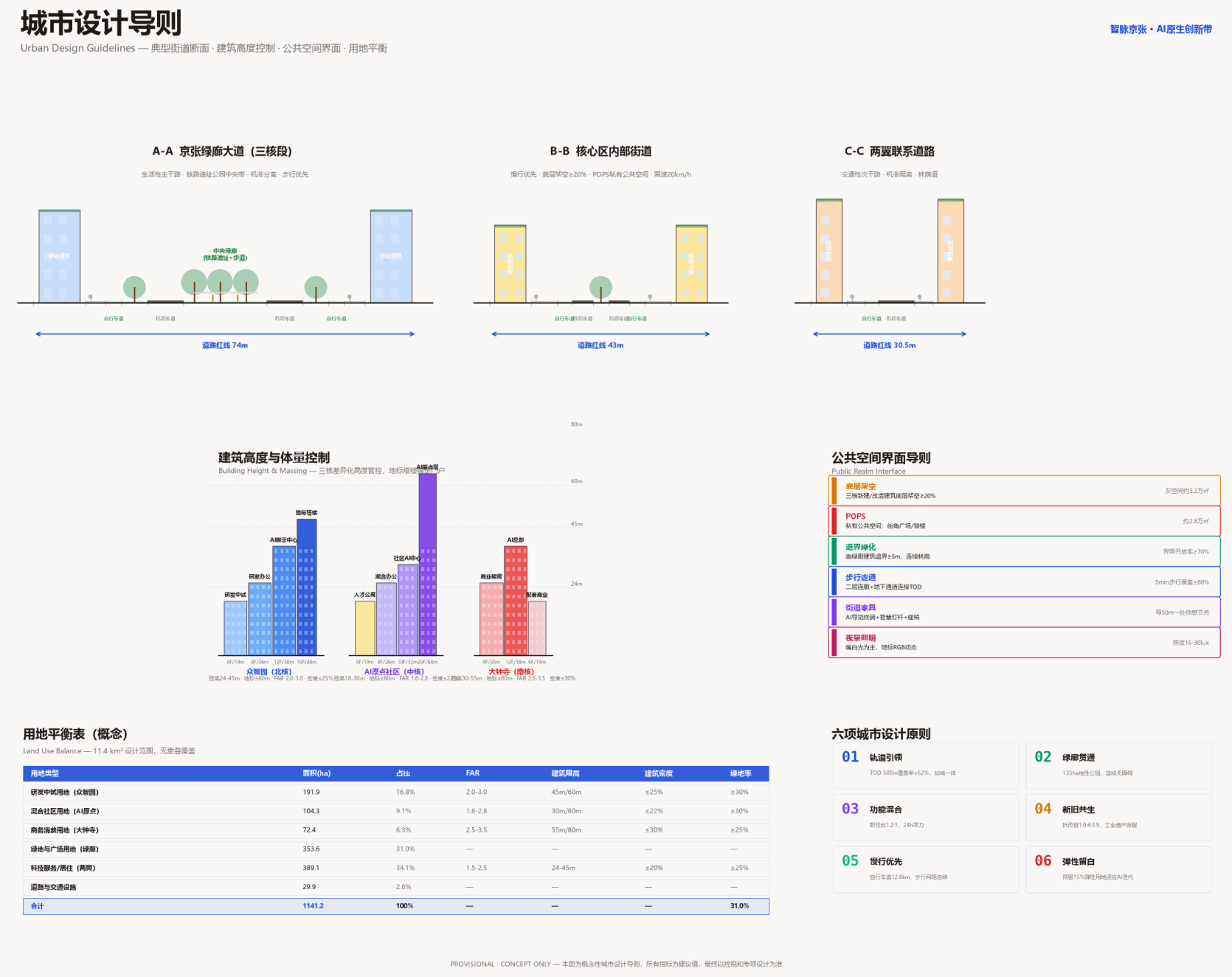


Fig.4b Urban Design Guidelines - Typical street sections, building height control, public space interface and land use balance (concept proposal)

Street Section: Jing-Zhang Green Corridor Avenue red line 74m, central 16m railway heritage park belt, motor/non-motor separation; core area internal street red line 43m, ground floor overhead $\geq 20\%$, POPS private public space, speed limit 20km/h; two wing connection road red line 30.5m, motor/non-motor isolated tree-lined road. **Height Control:** Three cores differentiated height control - Zhongzhi Park 24-45m (landmark $\leq 60\text{m}$), AI Origin 18-30m (landmark $\leq 60\text{m}$), Dazhongsi 30-55m (landmark $\leq 80\text{m}$). **Public Space:** Ground floor overhead grey space $\sim 32,000\text{m}^2$, POPS $\sim 28,000\text{m}^2$, setback from green corridor $\geq 5\text{m}$, 5min walking coverage $\geq 80\%$. All indicators are concept proposal values, subject to regulatory plan.

6. Regional Synergy and Full-Stack Independent Innovation

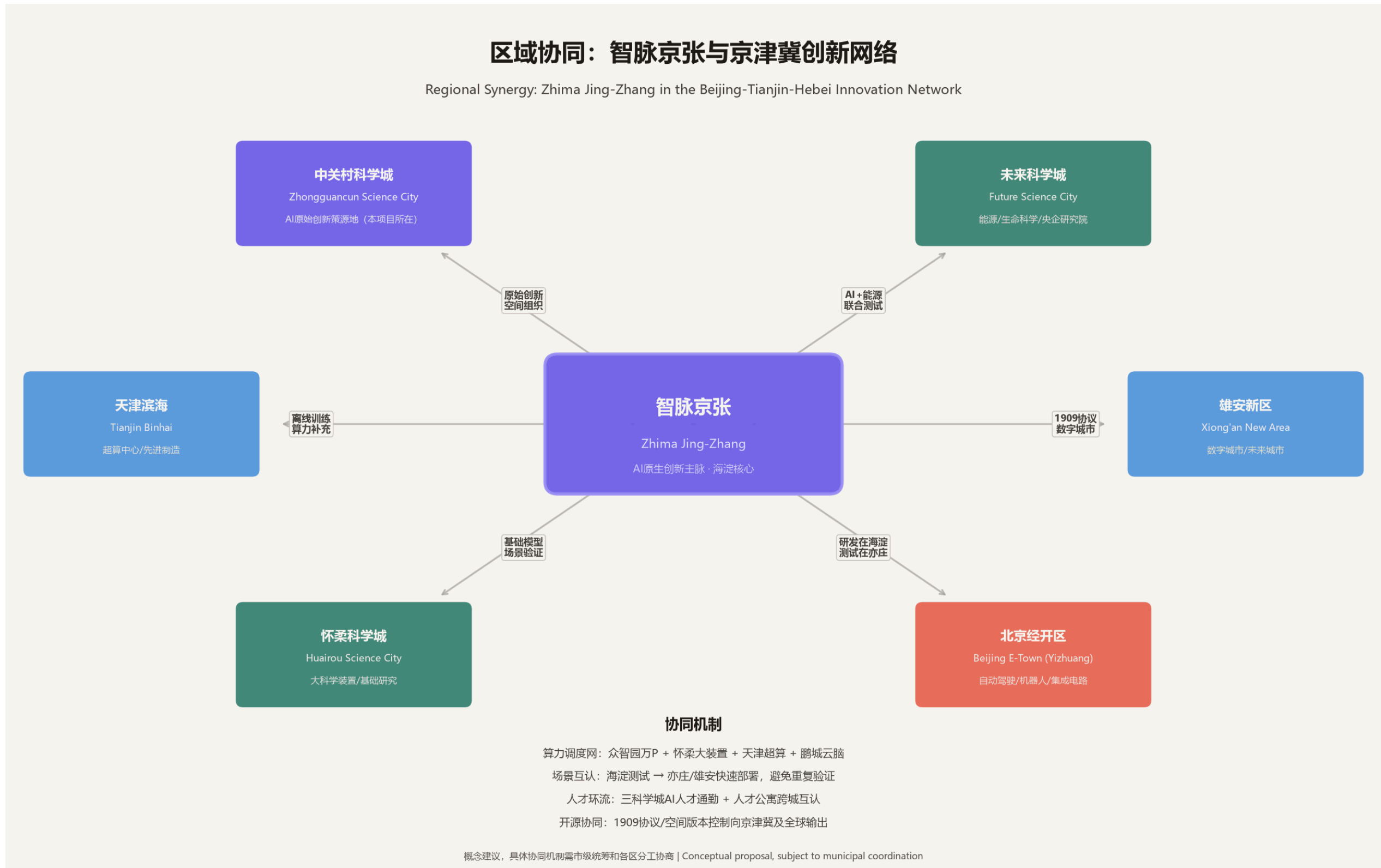


Fig.5 Regional Synergy - Zhimai Jing-Zhang with Three Science Cities, Economic Development Zone, Beijing-Tianjin-Hebei innovation network

Regional Synergy: Zhimai Jing-Zhang with Zhongguancun Science City (original innovation), Future Science City (energy/life science), Huairou Science City (large scientific facilities), Beijing Economic Development Zone (autonomous driving/robotics/chip manufacturing), Tianjin Binhai (supercomputing), Xiong'an New Area (digital city) form innovation network. Synergy mechanisms include computing power scheduling network, scenario mutual recognition, talent circulation and open source output. **Full-Stack Independent Innovation:** Zhongzhi Park layouts along 'chip -> framework -> model -> application' four-layer innovation chain.

7. Transit, Slow Traffic and Blue-Green Public Space

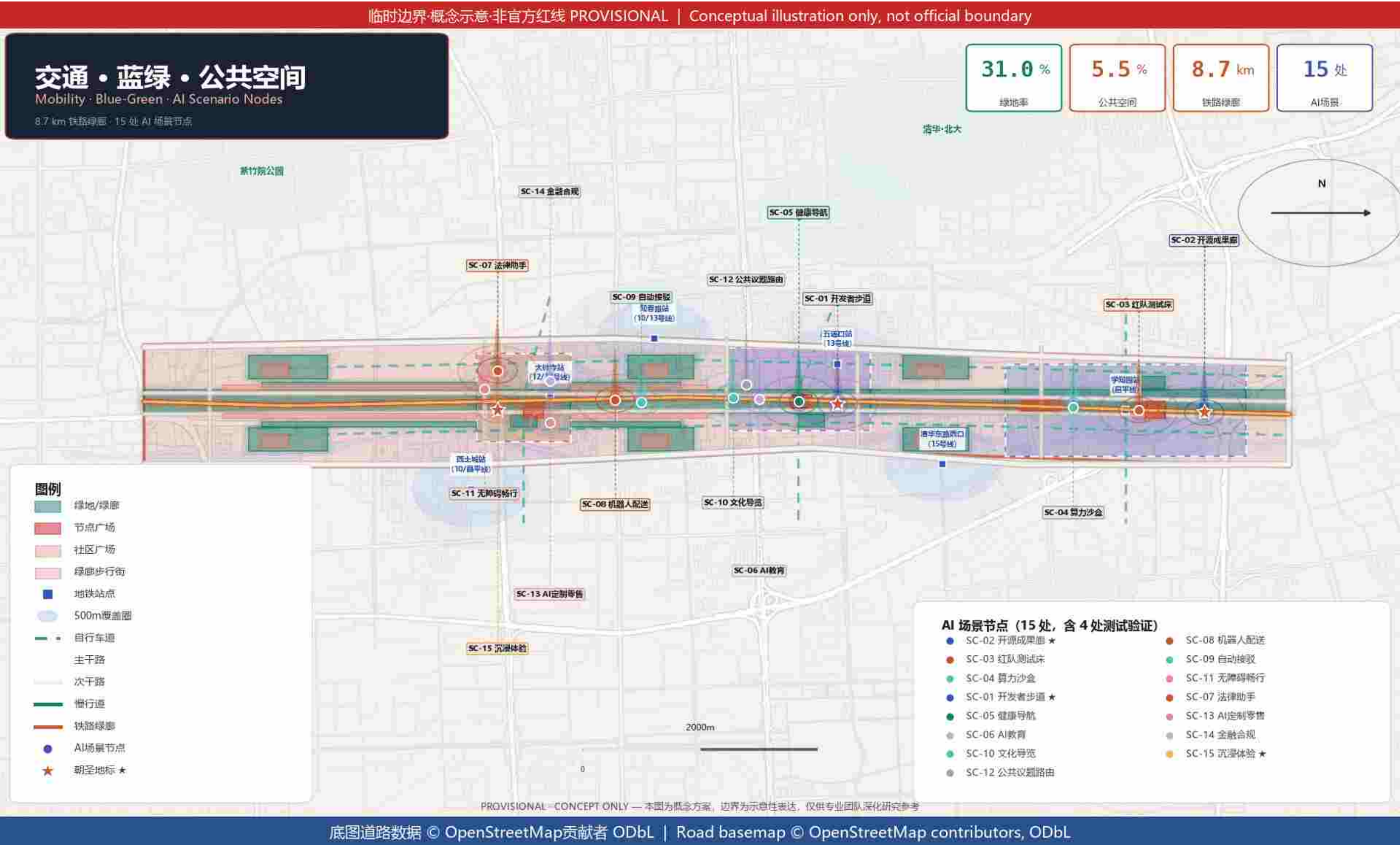


Fig.6 Transit, Slow Traffic and Blue-Green System - Railway Green Corridor + Road Grid + Public Space

8. AI+ Scenario Empowerment: 15 Scenario Cards

15 AI scenario cards cover research, entrepreneurship, education, health, legal, transit, culture, accessibility, public governance, commerce, finance, immersive experience and other dimensions. Among them, 4 are test verification scenarios (red team test bed, computing sandbox, robot delivery, autonomous driving shuttle), clearly marked as test status, not stated as approved operations. All scenarios follow data minimization, human-in-the-loop, test != operation three principles.

ID	Scenario Name	Target User	Status	ID	Scenario Name	Target User	Status
SC-01	Developer Promenade	Researchers/Entrepreneurs	Concept	SC-09	Autonomous Driving Shuttle	Commuters/Visitors	Test
SC-02	Open Source Gallery	Researchers/Students/Public	Concept	SC-10	AI Cultural Guide	Visitors/Citizens	Concept
SC-03	Model Red Team Test Bed	AI Safety Researchers	Test	SC-11	Accessibility Assistant	Disabled/Elderly	Concept
SC-04	Open Computing Sandbox	Startups/Researchers/Students	Test	SC-12	Public Issue Smart Routing	Residents/Merchants	Concept
SC-05	AI+ Health Navigation	Residents/Elderly	Concept	SC-13	AI Custom Retail Experience	Consumers/Brands	Concept
SC-06	AI Education Space	Students/Teachers	Concept	SC-14	AI Finance & Compliance Advisor	Entrepreneurs/SMEs	Concept
SC-07	AI Public Legal Assistant	Entrepreneurs/Residents	Concept	SC-15	AI Immersive Experience Space	Visitors/Families/Students	Concept
SC-08	Low-Speed Robot Delivery	Residents/Merchants	Test				

8 (Appendix). User Personas and Public Interest

5 Core User Types: AI Researchers (universities/research institutes, needs: computing, open data, peer exchange), AI Entrepreneurs (startups/SMEs, needs: low-cost space, test scenarios, capital), Students (university/graduate, needs: internships, learning, competitions), Community Residents (450k corridor residents, needs: convenience, privacy, not disturbed), Urban Governance (government/operator, needs: safety, manageable, auditable).

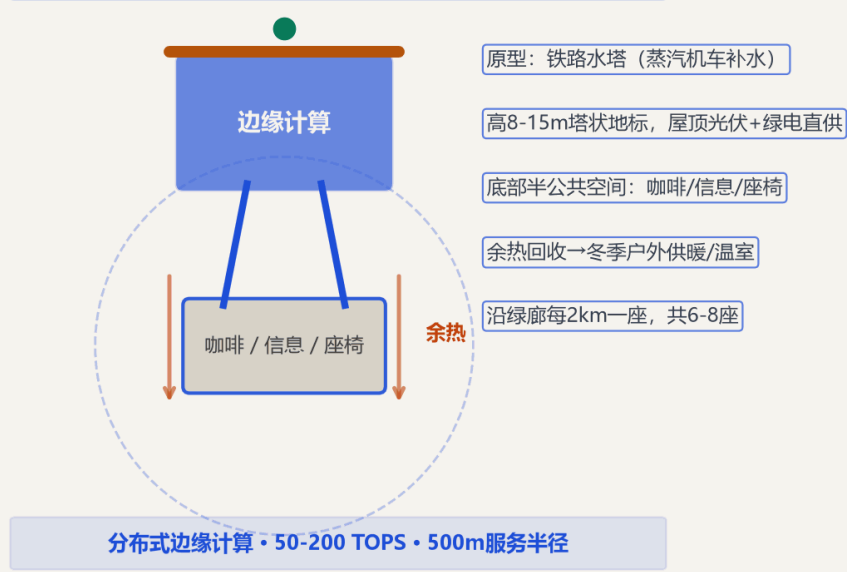
Public Interest Commitment	Specific Measures
Talent Housing >=20%	Rent not exceeding 80% of market rent, priority for Haidian employed talent
All-Age Friendly	All AI public services simultaneously provide non-digital channels (phone/window/paper), AI only assists not replaces
Anti-Gentrification	Original resident return ratio >=70%, community service facilities equal area replacement, rent increase exceeding 20% regional average triggers warning
Digital Inclusion	Annual digital literacy training >=1000 person-times, student computing free quota >=30%, accessibility design follows GB 50763

Three-Phase Community Participation: 1) Vision co-creation (0-12 months) workshops/surveys/interviews; 2) Co-construction supervision (1-3 years) plan review meetings/construction open days/digital dashboards; 3) Shared governance (3-10 years) community council quarterly voting/operation feedback/scenario priority co-decision. Scenarios with opposition rate exceeding 30% at community briefings are suspended and revised before re-consultation.

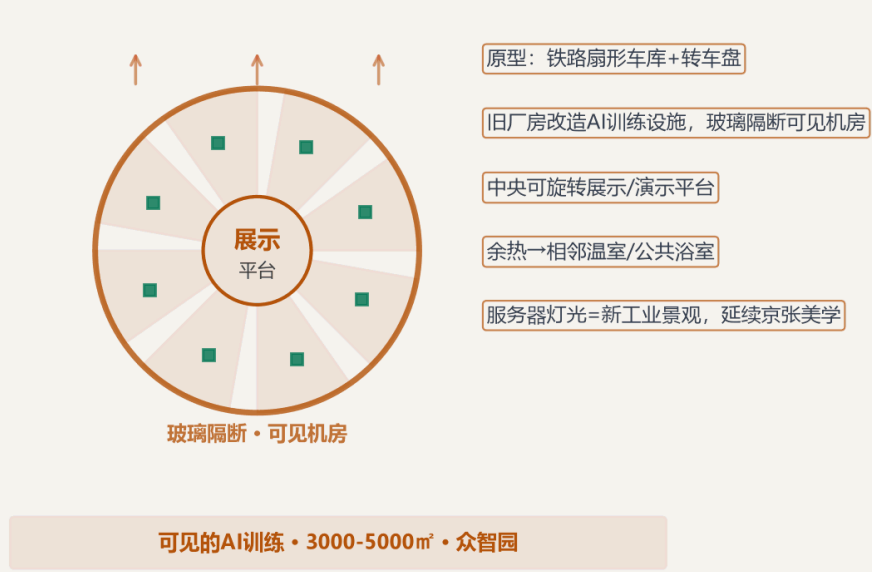
9. Original Mechanisms: 1909 Protocol and AI-Native Space

AI原生空间类型学 | AI-Native Spatial Typologies

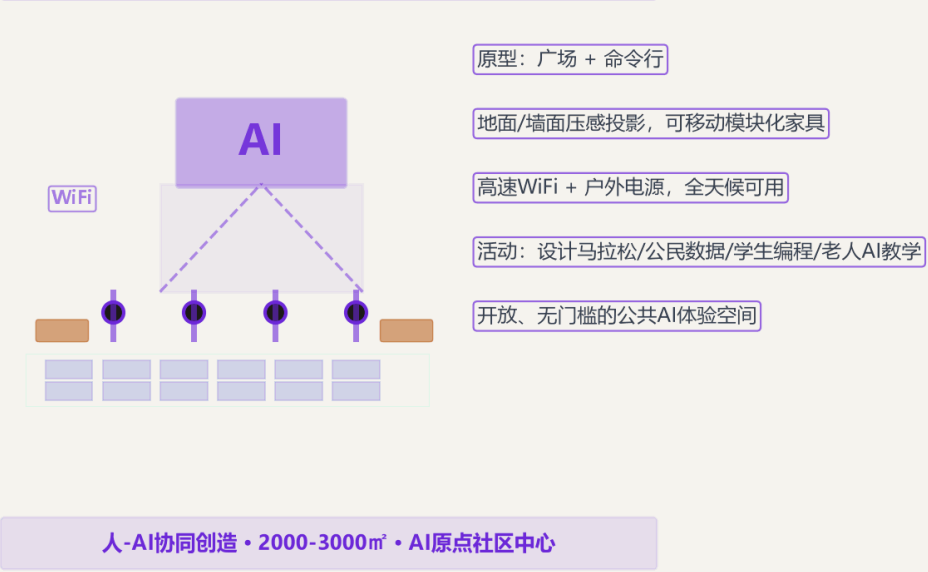
A 算力水塔
Compute Water Tower



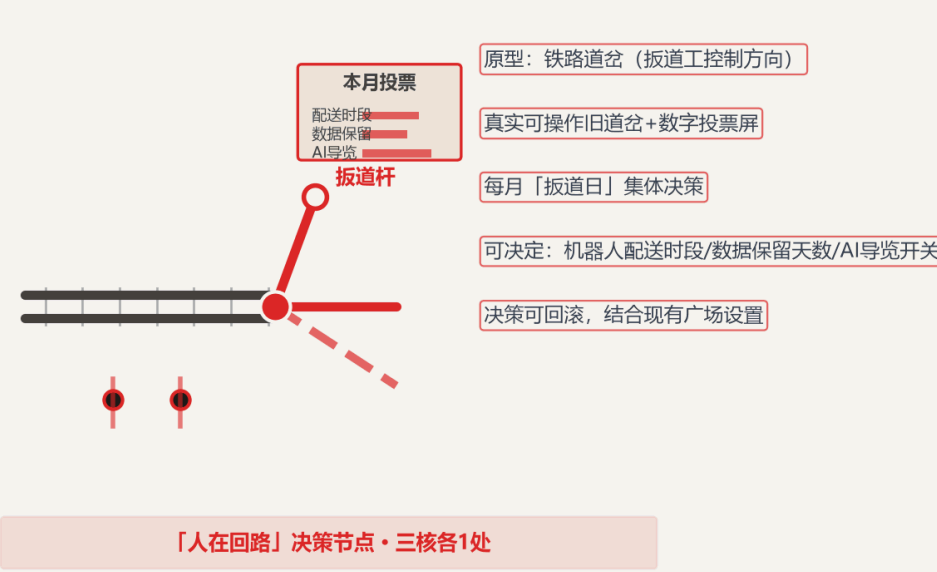
C 模型机房
Model Roundhouse



B 提示广场
Prompt Plaza



D 道岔广场
Switch Plaza



图注: 四种AI原生空间类型均为概念建议, 具体建筑设计、结构、设备需专业团队深化研究。

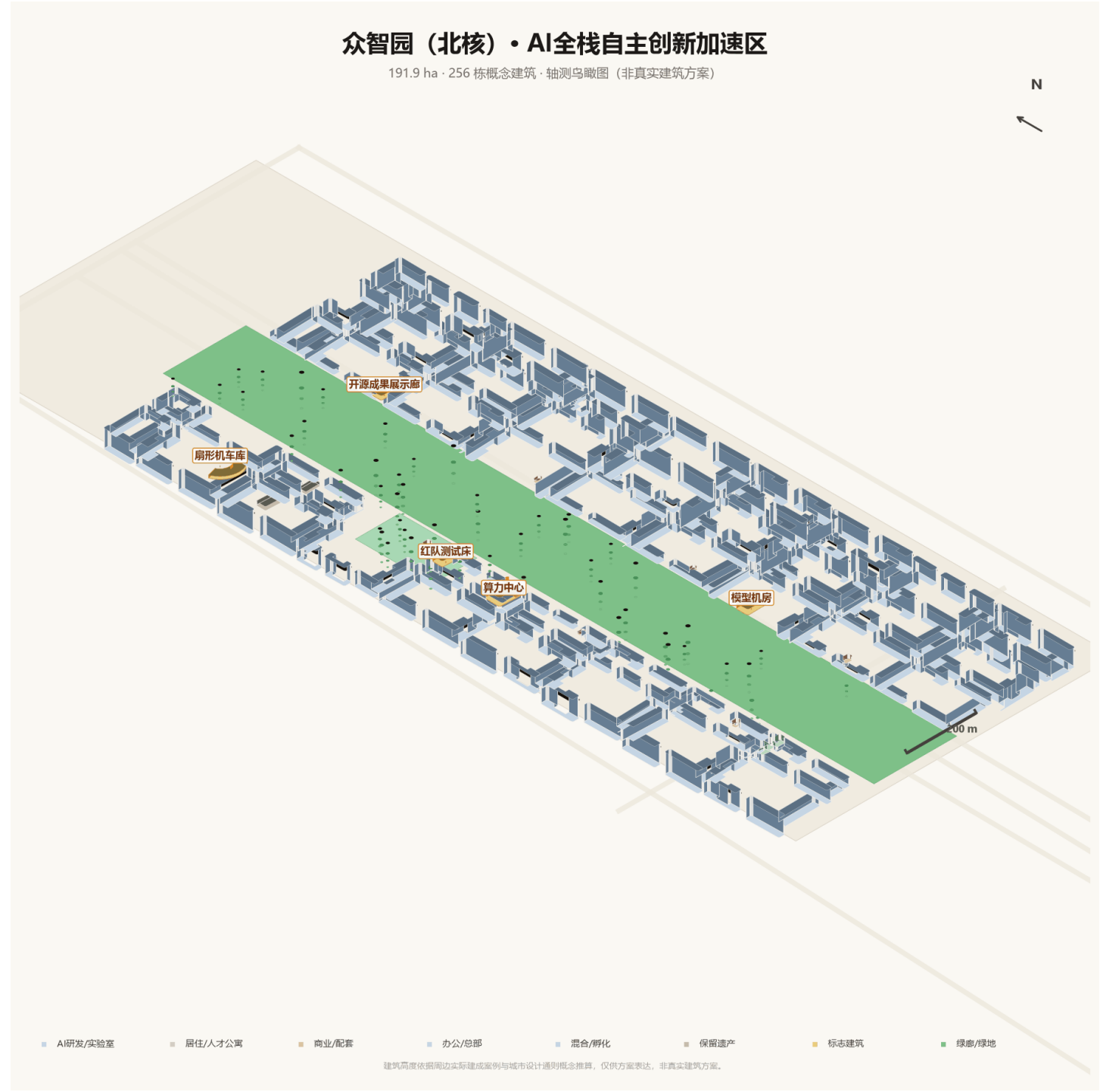
1909 Protocol: Based on the 1909 opening of Jing-Zhang Railway and railway signal block system as prototype, proposes interaction protocol between AI agents and urban space - block section (spatial sandbox), signal (state visible), interlocking (safety condition linkage), token (encrypted spatial credential), dispatch (human-in-the-loop approval), herringbone switchback (complex problem decomposition routing).

AI-Native Spatial Typology: Four new spatial types not existing in traditional urban design - A Computing Water Tower (edge computing landmark), B Prompt Plaza (human-AI co-creation), C Model Engine Room (visible training), D Switch Plaza (democratic decision-making).

Spatial Version Control: City OS supports AI configuration branch-test-merge-rollback (like git), community can trial-and-error, rollback, fork different AI configurations.

Zhimai City OS: Physical layer (neural green corridor) / Protocol layer (1909 protocol) / Governance layer (spatial version control). 'Zhimai Heartbeat' data light river continuously arranged along green corridor, ground LED + sensors, real-time display AI operation status.

9 (Appendix). Three Core Spatial Expression (1): Zhongzhi Park (North Core)



Axonometric Bird's Eye View - 191.9ha - 256 concept buildings

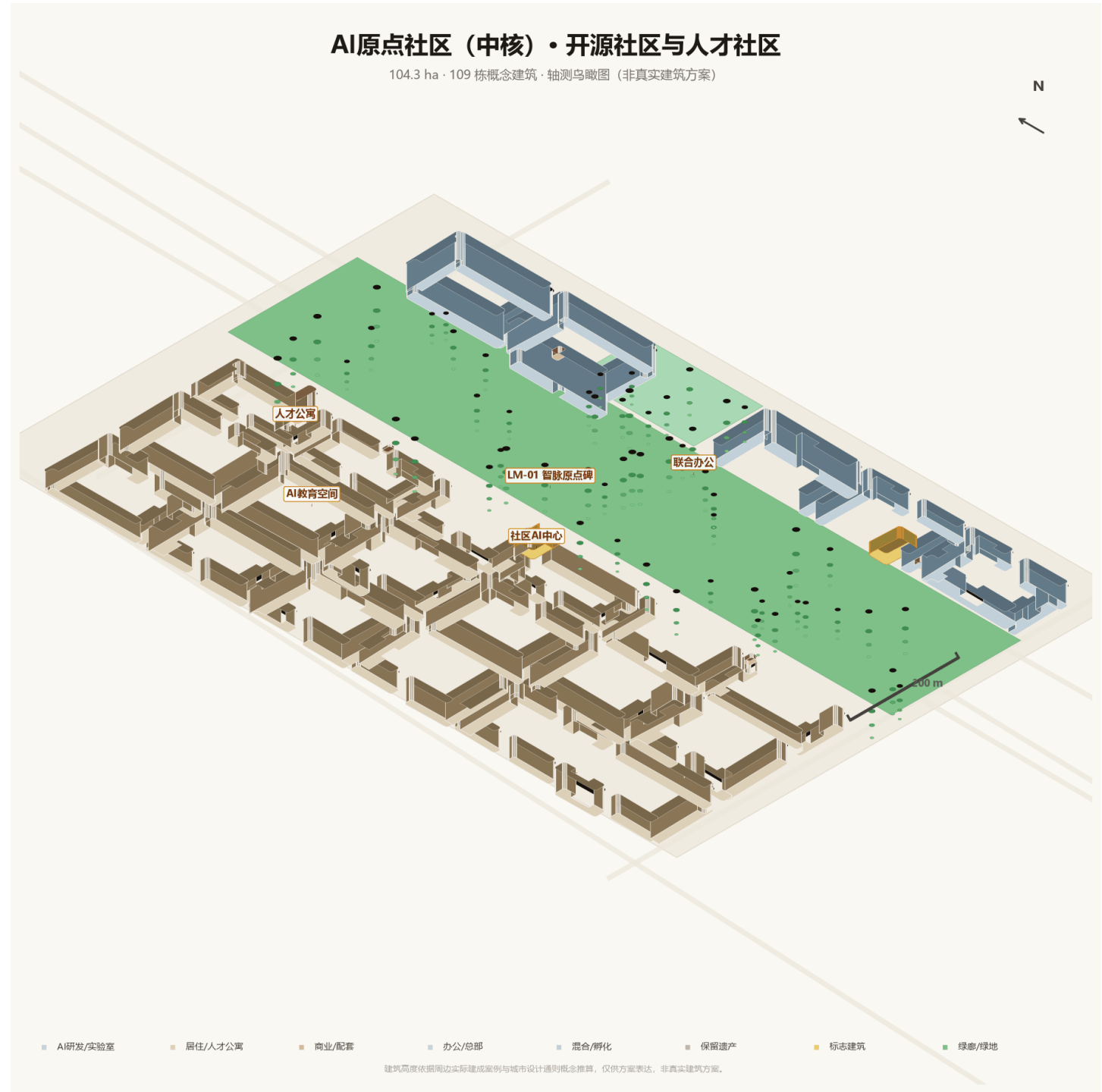
AI full-stack independent innovation acceleration zone: large model training pilot base, red team test bed (SC-03), open computing sandbox (SC-04), open source achievement gallery (LM-02). Preserving fan-shaped locomotive depot industrial heritage, dark computing center + glass R&D; building cluster, linear green corridor north-south through.

Left: data-driven axonometric analysis map (based on buildings.geojson actual coordinates and floor counts, building height conceptually calculated as floors x 3.5m); Right: AI-generated rendering intention, expressing spatial atmosphere and materials, building count, street grid and specific positions do not correspond to real map, spatial positions subject to axonometric analysis map and master plan. provisional boundary, official_boundary=false.



Rendering Intention - Fan-shaped locomotive depot transformed into open source gallery

9 (Appendix). Three Core Spatial Expression (2): AI Origin Community (Central Core)



Axonometric Bird's Eye View - 104.3ha - 109 concept buildings

24h mixed innovation community: warm-color residential 6-12F, blue-glass mixed tower up to 20F (TOD node), talent housing >=20%. Central green corridor children's playground, Zhimai Origin Monument (LM-01) at community central plaza. Residential, entrepreneurship, education, consumption mixed layout.

Left: data-driven axonometric analysis map (based on buildings.geojson actual coordinates and floor counts, building height conceptually calculated as floors x 3.5m); Right: AI-generated rendering intention, expressing spatial atmosphere and materials, building count, street grid and specific positions do not correspond to real map, spatial positions subject to axonometric analysis map and master plan. provisional boundary, official_boundary=false.



Rendering Intention - Residential/talent housing and TOD mixed tower

9 (Appendix). Three Core Spatial Expression (3): Dazhongsi (South Core)



Axonometric Bird's Eye View - 72.4ha - 76 concept buildings

AI industry gateway: Golden AI headquarters building 26F ~91m as tallest in proposal, blue-grey office tower cluster, red commercial district. Global AI Milestone Pavilion (LM-04) at gateway plaza, transit transfer hub, green corridor north-south through without enclosure.

Left: data-driven axonometric analysis map (based on buildings.geojson actual coordinates and floor counts, building height conceptually calculated as floors x 3.5m); Right: AI-generated rendering intention, expressing spatial atmosphere and materials, building count, street grid and specific positions do not correspond to real map, spatial positions subject to axonometric analysis map and master plan. provisional boundary, official_boundary=false.



Rendering Intention - Golden AI headquarters building (26F ~91m)

10. Cultural Narrative: From Herringbone to Network

Layer 1 - Century-old Jing-Zhang

In 1909, Zhan Tianyou presided over construction of China's first railway, the 'herringbone' line solving the Guangou slope problem. From 'herringbone' to 'network', Zhan Tianyou's switchback wisdom metaphorizes AI red-teaming and continuous iteration. Railway signal block system evolves into '1909 Protocol' spatial interaction rules.

Layer 2 - Zhongguancun

From electronics street to world's highest AI density area - 1900+ AI enterprises, 51 unicorns, 104 large models, 646 academicians. Innovation iteration spirit on the same land, from hardware trade to algorithm open source, from 'China's Silicon Valley' to 'AI-native community'.

Layer 3 - AI New Culture

Open source collaboration (code is road, commit is footprint), rapid iteration (spatial version control), human-machine symbiosis (AI doesn't replace but amplifies humans), ethical consciousness (red team test bed and AI safety one-vote veto), cross-domain integration (AI+culture/health/education/legal).

Spatial Storyline along main artery from north to south: Open Source Gallery (today's innovation, tomorrow's heritage) -> Developer Promenade (code is road, commit is footprint) -> Zhimai Origin Monument (from herringbone to network) -> Wudaokou Stitching Plaza (level crossing -> digital intersection) -> Global AI Milestone Pavilion (from railway to computing, from China to world).

Four AI Pilgrimage Landmarks

ID	Name	Location	Concept
LM-01	Zhimai Origin Monument	AI Origin Community Central Plaza	Marks AI-native innovation starting point, with open source code time capsule
LM-02	Open Source Achievement Gallery	Zhongzhi Park entrance old factory	China open source AI achievement hall of honor, fan-shaped locomotive depot transformation
LM-03	Developer Promenade	Middle section of green corridor	Outstanding developer code starlight, ground carved commit records
LM-04	Global AI Milestone Pavilion	Dazhongsi gateway plaza	Global AI development node display, from Turing to deep learning

Indicator	Recalculated Value	Confidence	Indicator	Recalculated Value	Confidence
site_area_sqm	11.41 km²	medium	ai_pilgrimage_landmarks	4	high
research_area_sqm	43.61 km²	medium	talent_apartment_ratio	20.0%	low
key_area_total_sqm	3.69 km²	medium	investment_estimate_total_yi	150	low
land_use_total_sqm	11.41 km²	medium	urban_os_phased_years	10	low
green_ratio	31.0%	medium	public_benefit_categories	6	high
public_space_ratio	5.5%	medium	building_count	441	low
building_footprint_area_sqm	65.4 10k m²	low	transit_accessibility_index	72.0%	medium
ai_scenario_nodes	15	high	road_network_density_km_per_sqkm	4.8	medium
key_areas_count	3	high			



12. Event Operations and Long-Term Mechanisms

Event	Frequency	Time	Core Content	Target Audience
Zhimai Open Week	Annual	September (Jing-Zhang opening ceremony)	Open source release - Hackathon finals - AI art exhibition - Public day	Global developers/Public
AI Force Summit	Annual	Spring	Keynote speeches - Trend releases - Investment matchmaking	Industry/Academia/Capital
Open Source Hackathon	Quarterly	First month each quarter	48h collaboration - Mentor guidance - Prize pool	Developers/Students
Red Team Challenge	Semi-annual	April/October	AI security attack-defense - Vulnerability discovery - Ethics review	Security researchers
AI Art Festival	Annual	Autumn	AI-generated art - Human-machine co-creation - Outdoor projection	Public/Artists
Public AI Day	Monthly	First Saturday each month	AI experience - Science lectures - Community workshops	Residents/Families

International Communication

- Chinese-English bilingual core materials, collaboration with GitHub/Hugging Face
- Overseas developer community connection, international media and academic channel output
- 'Railway heritage + AI-native' narrative distinguishes from pure tech parks
- Annual Open Week invites global AI lab representatives

Developer Community Five Mechanisms

- Residency program: 3-6 months free workspace + computing
- Mentor network: academicians/unicorn CTOs/open source leaders
- Open source bounty: key issue rewards
- Community governance: developer committee participates in spatial decisions
- Knowledge■■■: public tech sharing and documentation

Attraction Conversion: 1-on-1 matchmaking within 2 weeks after Open Week -> landing incentive package within 1 month (space/computing/talent/capital/scenario) -> settlement decision within 3 months, connecting with Zhongguancun policies.

Phase	Time	Investment Estimate	Funding Structure	Key Projects
Near-term	1-3 years	1.5-2.5B yuan	Government guidance mainly	Zhimai Origin Plaza - City OS Phase 1 - AI scenario test environment
Mid-term	3-5 years	4-6B yuan	Government guidance + social capital	Zhongzhi Park shared pilot layer - Computing sandbox - Talent housing completion
Long-term	5-10 years	6-10B yuan	Social capital mainly	Dazhongsi AI industry gateway - Two wings full operation - City OS iteration
Total	10 years	11.5-18.5B yuan	-	Median ~15B yuan (conceptual magnitude, not investment commitment)

13. Implementation Path: Current Status Analysis

Fig.8 Three Core Current Status Analysis - Preservation areas, renewable plot concept classification, real institutional anchors (conceptual, subject to ownership survey)

Renewable Land Concept Judgment: Within three cores, large areas are built university campuses and research institutes (green preservation areas), renewable plots mainly concentrated along railway, old factories, some commercial offices (golden areas), specific scope subject to current building survey, land ownership investigation and regulatory plan adjustment confirmation.

13. Implementation Path: Roadmap and Governance

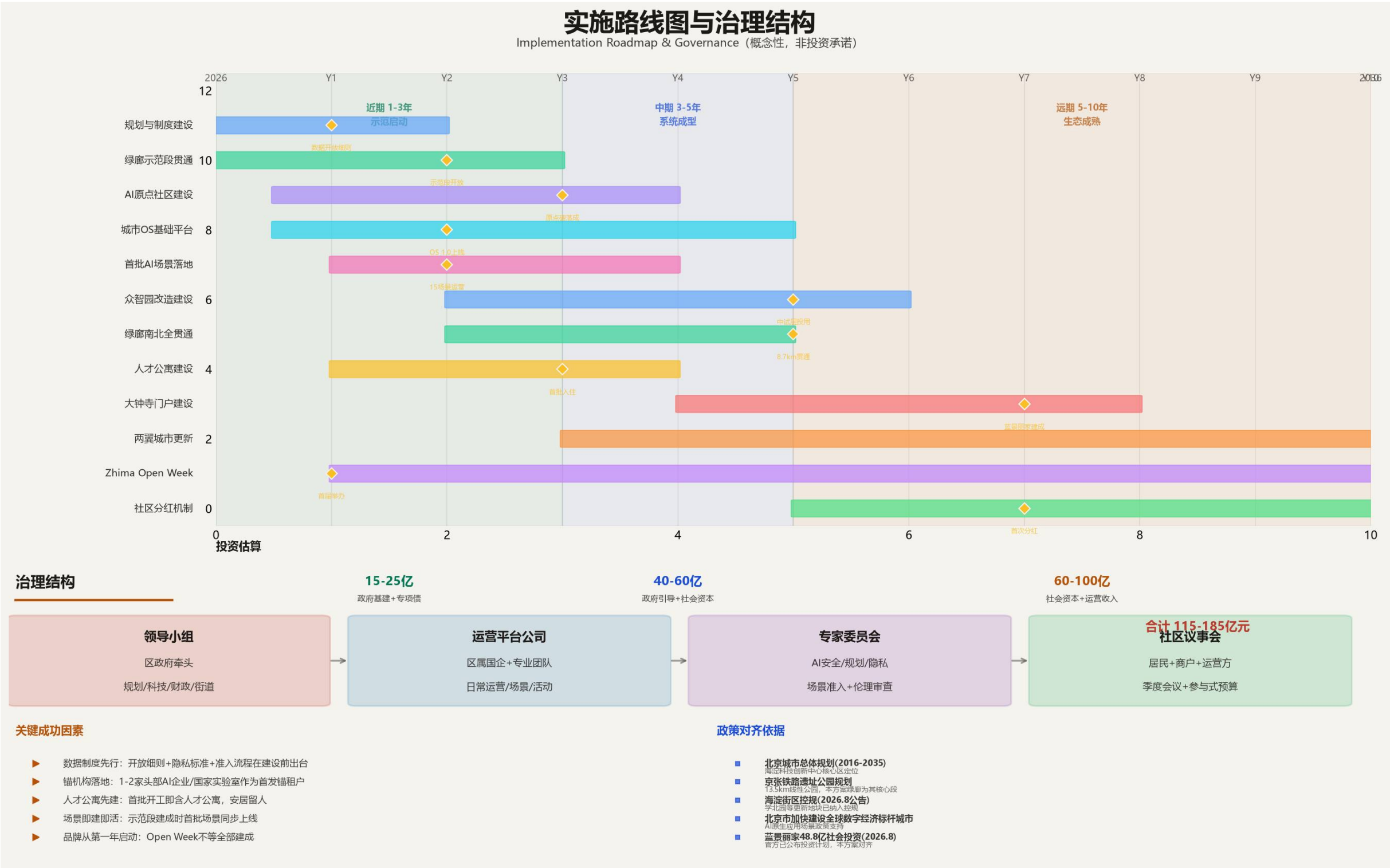


Fig.9 Implementation Roadmap - Near-term 6 major projects (N-01 to N-06, 1.5-2.5B yuan), phased investment, governance structure (not investment commitment)

Governance Structure: Leadership group (district government) + Operation platform company (SOE + professional team) + Expert committee (access and ethics, one-vote veto) + Community council (participatory budgeting). Investment figures are conceptual magnitude, not investment commitment, subject to feasibility study.

13 (Appendix). Risk Matrix and Responsibility Division

Risk Category	Level	Core Mitigation Measures
Data Privacy and Personal Information Protection	2	Minimal necessary collection - Sensors do not collect faces/license plates - Data localization - Annual third-party audit
Implementation Complexity and Multi-stakeholder Coordination	2	District leadership group monthly scheduling - PMO milestone management - Unclear ownership not included in first phase - 10% contingency fee
Public Acceptance and Community Identity	3	Preserve railway heritage - Three-phase participation - Preserve existing usage habits - Community AI experience day
Long-term Operation Cost and Sustainability	4	Basic operation included in finance (30-50M/year) - Value-added income - Year 5 government input <=30%
Policy Uncertainty and Approval Risk	4	All marked concept proposal - Policy quarterly tracking - Test use regulatory sandbox - Reserve policy elasticity
Spatial Dispute and Interest Balance	3	Three core differentiated positioning - Public space only increases not decreases - Community service facilities equal area replacement
Technology Maturity and AI Reliability	3	Three gates + eight fault injection types (960 tests) + four-level degradation - Synthetic != field - Five-phase verification
Fairness and Digital Divide	3	Preserve non-digital channels - Large print/voice/multilingual - Annual training >=1000 person-times - GB 50763

Risk Level Description: 1=Very low - 2=Low - 3=Medium (yellow) - 4=High (red, requires manual review) - 5=Very high

Suggested Responsibility Division Framework (not government arrangement, final subject to government decision)

Work Area	Suggested Lead	Key Responsibilities
Planning and Land	District Planning Bureau	Concept plan deepening - Regulatory plan connection - Land preparation
Industry and AI Ecosystem	District Science & Tech Bureau	AI enterprise attraction - Computing infrastructure - Policy connection
Infrastructure and New Infrastructure	District Development & Reform Bureau	5G/IoT - City OS - Energy system
Public Space and Green Corridor	District Landscape Bureau	Green corridor construction maintenance - Pocket parks - Public art
Urban Renewal and Construction	Urban renewal platform/district SOE	Old factory transformation - Talent housing - TOD development
Operations and Events	Operation platform company	Open Week - Daily operations - Spatial version management
Community and Public Participation	Subdistrict office	Three-phase participation - Community AI day - Feedback
AI Safety and Ethics	Expert committee (one-vote veto)	Red team testing - Ethics review - Data governance supervision
International Communication and Brand	Operation platform brand dept	Bilingual materials - Overseas collaboration - Brand IP management

13 (Appendix 2). AI Verification Protocol and Fault Injection Testing

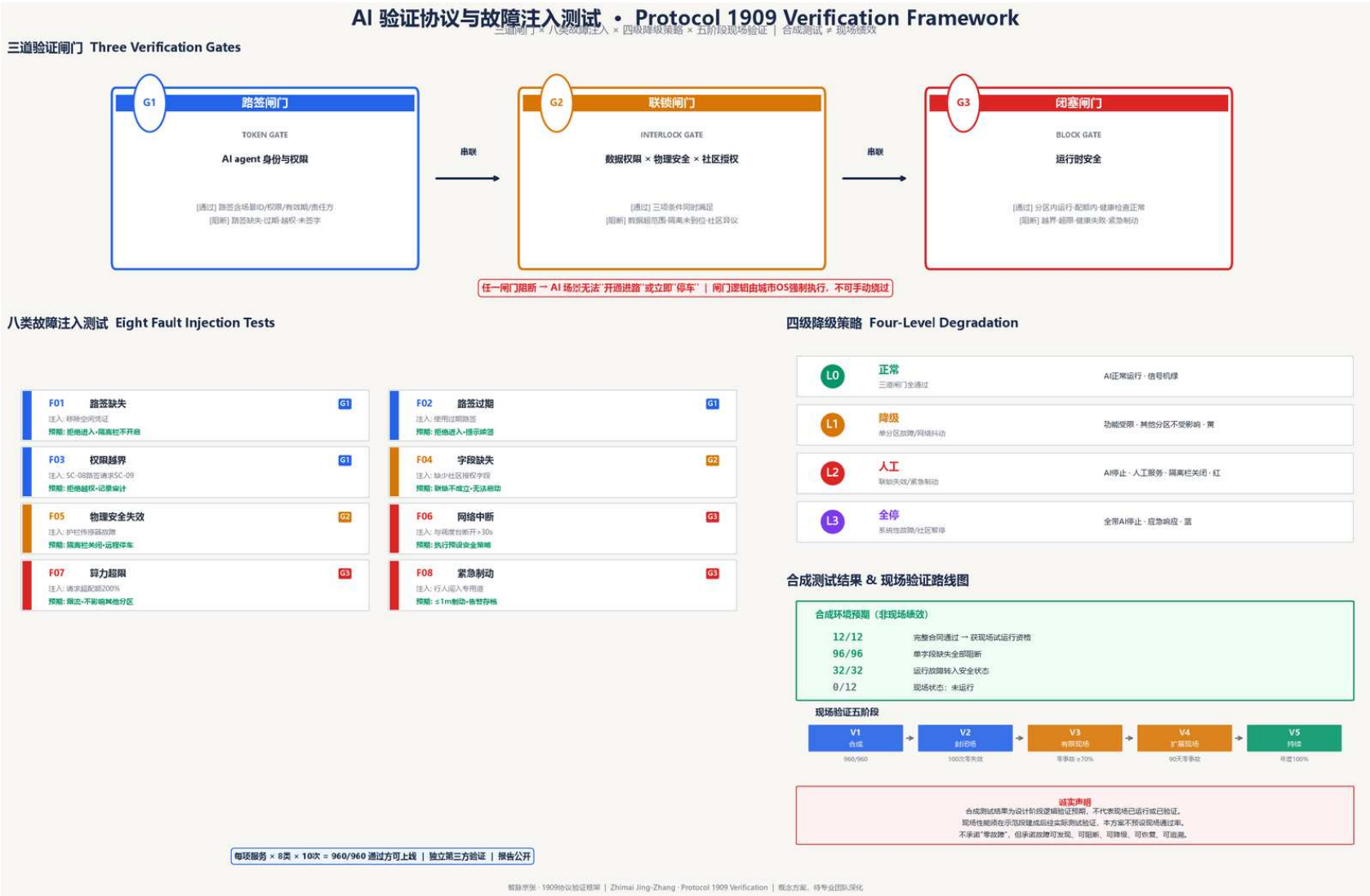


Fig.10 AI Verification Protocol - Three gates (token/interlocking/block), eight fault injection types, four-level degradation strategy, five-phase field verification (synthetic test != field performance)

Verification Framework: 1909 Protocol corresponds to three verification gates - G1 Token (identity permission), G2 Interlocking (data x physical x community), G3 Block (runtime safety), any gate blocked means AI scenario cannot start or immediately stops. **Fault Injection:** Each service executes 8 types x 10 times = 960 tests in sandbox before launch, 10/10 pass required for launch. **Degradation Strategy:** L0 normal -> L1 degraded -> L2 manual -> L3 full stop, manual service response <=5 minutes. **Honesty Declaration:** Synthetic test results (12/12, 96/96, 32/32) are design phase logical expectations, field status 0/12 not running, no field pass rate preset.

14. Risk and Compliance Declaration

This proposal strictly follows the following boundaries: No regulatory plan adjustment/FAR/building height/intensity conclusions; No specific plot demolition/retain; No road alignment/rail alignment/bridge-tunnel/municipal pipeline engineering; No underground space engineering feasibility/energy load/municipal capacity; No land ownership/investment commitment/development schedule approval judgment; No non-public data; No concept proposal stated as determined decision. All spatial proposals are 'concept proposal/reference plan/for professional team further study'. All provisional geometry: official_boundary=false, not official red line. Base maps all self-drawn, no third-party commercial map tiles; Fonts use system fonts; Images all self-drawn or AI-generated. Test verification scenarios (red team/sandbox/delivery/shuttle) clearly marked as test status, not stated as approved operations.

Data Governance Five Principles Implementation Methods

Principle	Implementation Method
Minimal Necessary	Sensors do not collect faces/license plates, only anonymous■■■/environmental data
Local Processing	Edge computing node local processing, raw data does not leave park
Classification Grading	Public/internal/sensitive three levels, sensitive data encrypted storage
Transparent Auditable	Data processing log public, annual third-party audit
Human-in-the-loop	Key decisions require manual confirmation, AI only provides auxiliary suggestions

Compliance Self-Check List

Compliance Item	Status
All spatial proposals marked 'concept proposal/reference plan'	Met
provisional boundary official_boundary=false	Met
No regulatory plan adjustment/FAR/building height conclusions	Met
No specific demolition/retain/road alignment/bridge-tunnel engineering	Met
No investment commitment/non-public data	Met
HTML offline openable, no CDN/external dependencies	Met
Base map self-drawn, font system font, image self-drawn/AI-generated	Met
Test verification scenarios marked test status	Met
A3/A0 PDF each <10MB, assets single file <=5MB	Met

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